

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit: Geology and Geophysics**
2. **Subject Name: Remote Sensing and GIS Laboratory L-T-P: 0-0-3**
Credits:
3. **Pre-requisites: Remote Sensing and GIS theory**
4. **Syllabus and reference books:**

Syllabus:

Interpretation of aerial photographs and satellite images; study of fluvial, karst, coastal, aeolian landforms; preparation of thematic maps from aerial photographs and satellite images; photogrammetric computations involving parallax and vertical exaggeration; determination of heights of objects using parallax bar; morphometric analyses involving remote sensing data.

Digital Remote Sensing: geometric rectification, various image processing techniques; supervised and unsupervised classification of satellite image; generation of various thematic maps.

GIS database study and query, digitisation in ArcGIS.

Reference Books:

- 1) Encounter Earth: Interactive Geoscience Explorations by Kluge
- 2) ERDAS Field guide.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Quantitative interpretation of satellite image	5
2.	Digital image processing in Excel	5
3.	Introduction of ERDAS and ArcGIS	5
4.		
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Total number of hours		45